

Project 1: Data Cleaning & Preparation

Change Log & Audit Report

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Dataset: Dataset_for_Data_Analytics.xlsx
Original Records: 1,200 | Final Records: 1,200

1. Executive Summary

The raw dataset was audited end-to-end for missing values, duplicate records, and formatting inconsistencies across identifiers, dates, currency fields, and categorical text columns. One genuine data quality issue was found and resolved (missing coupon codes). All other integrity checks — duplicate Order IDs, duplicate rows, date format validity, and price calculation consistency — passed with a 0% error rate, meeting the verification threshold required to proceed.

2. Change Register

Change ID	Description	Impact / Finding	Status
CR001	Missing values in 'CouponCode' column	309 of 1,200 records (25.75%) had no coupon code. Imputed as the explicit category 'No Coupon' rather than mean/median/mode, since the absence reflects a real business state (customer used no coupon), not a random data-entry gap.	Resolved
CR002	Duplicate record audit (Order IDs & full rows)	Checked all 1,200 Order IDs for uniqueness and all rows for exact duplication. 0 duplicate Order IDs and 0 duplicate rows found. No removal needed.	Verified — Pass
CR003	Date format standardization ('Date' column)	All 1,200 date values were already valid calendar dates. Standardized display format to ISO 8601 (YYYY-MM-DD) for consistency. 0 invalid/malformed dates found.	Verified — Pass
CR004	Numeric precision ('UnitPrice', 'TotalPrice')	Enforced 2-decimal precision on all currency fields. Recomputed Quantity x UnitPrice and cross-checked against TotalPrice for all rows: 0 mismatches found.	Verified — Pass
CR005	Text formatting (IDs & categorical fields)	Trimmed whitespace and standardized casing on OrderID, CustomerID, TrackingNumber, and categorical fields (Product, PaymentMethod, OrderStatus, ReferralSource, CouponCode). No casing/whitespace defects were found; step applied as a safeguard.	Verified — Pass

3. Verification Gate — Threshold for Project 2

Check	Requirement	Result
Unique Identifiers	0% error rate on duplicate Order IDs	0% — PASS
Date Formats	0% error rate on incorrectly formatted dates	0% — PASS

4. Tools & Method

Cleaning was performed using Python (pandas) for a reproducible, scriptable workflow, as recommended for datasets of this scale. The process: (1) load raw data, (2) audit for duplicates and nulls, (3) impute missing categorical values with an explicit label, (4) standardize date and numeric formats, (5) trim/normalize text fields, (6) cross-validate calculated fields, (7) export the cleaned dataset.